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Uncertainty Estimation in Reservoir Property Prediction Using Machine Learning: A Gulf of Mexico Case Study

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Abstract

Accurate prediction of reservoir properties is essential for reducing exploration and development risks in subsurface resource evaluation. Machine learning (ML) techniques, particularly deep learning, have demonstrated strong capabilities in mapping reservoir properties from seismic data. However, the reliability of such predictions remains a critical concern, especially in data-scarce environments. This study addresses this challenge by focusing on uncertainty estimation in ML-based reservoir property prediction, using a case study from the Gulf of Mexico.

A rock physics-guided convolutional neural network (CNN) approach was used, with the CNN trained on a synthetic well catalog. The catalog was generated by calibrating a rock physics model based on four real wells, with a fifth well reserved entirely for blind testing. Scenario modeling was then applied to capture a wide range of geological variations, including changes in reservoir thickness, porosity, and water saturation. For each pseudo-well, synthetic seismic gathers were simulated, resulting in a comprehensive training dataset for predicting P-impedance, porosity, and water saturation.

To evaluate the model's robustness and quantify uncertainty, the jackknife validation technique was applied. In this approach, one real well was excluded from the training set in each iteration, enabling the creation of multiple model realizations. This allowed the computation of statistical measures, such as the mean and standard deviation of predictions, offering insights into prediction confidence.

Following the initial training, transfer learning was used to fine-tune the CNN using real seismic data and four real wells. The CNN-predicted property volumes showed strong agreement with filtered log curves, including at a blind well, demonstrating the model's generalization capabilities.

Uncertainty maps revealed that areas in the central and northwestern regions of the study zone exhibit high porosity and low water saturation with correspondingly low uncertainty, indicating high confidence in predictions. Conversely, areas with high uncertainty – particularly in the southeast – highlight possible influences of seismic noise or complex geology.

This integrated approach, combining synthetic data generation, rock physics modeling, machine learning, transfer learning, and jackknife-based uncertainty quantification, offers a robust framework for reservoir characterization. It is especially valuable in regions with limited well control, providing not only accurate predictions but also actionable measures of confidence for decision-making.

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Introduction

Seismic inversion is a widely used technique for predicting reservoir properties, providing valuable insights into subsurface characteristics. Geostatistical seismic inversion, in particular, not only delivers high-resolution results but also enables uncertainty quantification, which is essential for risk-aware decision-making. However, several challenges limit its applicability and reliability in certain scenarios.

One major challenge arises when well control is sparse. In such cases, the lack of sufficient calibration data undermines the inversion's ability to capture geological variability, particularly in complex and heterogeneous environments. Another limitation of the most traditional and commonly used geostatistical inversion approaches is the indirect prediction workflow, where elastic properties are first estimated and subsequently transformed into reservoir properties. This multi-step process can introduce error accumulation and reduce the accuracy of predicting target variables such as porosity or water saturation. Although direct inversion approaches to reservoir properties do exist, they remain less commonly implemented in standard workflows.

ML, and deep neural networks (DNNs) in particular, offer a promising alternative by enabling direct prediction of target reservoir properties from seismic data. These models have shown potential to outperform traditional methods in terms of accuracy and flexibility. However, their performance is highly dependent on the availability of large, labeled datasets. Training deep networks with limited real well data increases the risk of overfitting and reduces the generalization capacity of the model.

To overcome this limitation, we used a methodology that leverages rock physics modeling and scenario simulation to generate a synthetic catalog (Downton et al., 2020). This catalog captures a wide range of geological scenarios and provides a rich training set for ML models, even in data-scarce environments. A CNN is first trained on this synthetic data and then adapted to real field data using transfer learning, allowing the model to account for field-specific seismic characteristics and noise.

Crucially, this approach also addresses the need for uncertainty quantification. By applying the jackknife validation technique, where individual wells are systematically excluded from training to generate multiple model realizations, we estimate prediction variability and generate confidence maps alongside predicted property volumes. This integrated framework enhances the reliability of ML-based reservoir characterization and supports informed decision-making in uncertain geological settings.

Statement of Theory and Definitions

Modern reservoir characterization increasingly relies on the integration of data-driven approaches with established physical models to improve the accuracy and reliability of subsurface predictions. This section outlines the core theoretical components and definitions that form the foundation of the presented methodology, which combines machine learning, rock physics, synthetic data simulation, and uncertainty quantification. The concepts detailed below reflect a hybrid approach that leverages both physical interpretability and statistical learning.

Reservoir Property Prediction

The goal of reservoir property prediction is to infer subsurface petrophysical parameters – such as porosity and water saturation – from seismic and well log data. Traditional approaches rely on seismic inversion followed by empirical or statistical transformations to obtain reservoir properties. While effective, this two-step process may suffer from accumulated errors and lacks flexibility in handling complex geological settings (Hampson et al., 2001).

Convolutional Neural Networks (CNNs)

CNNs are a class of deep learning models particularly effective in capturing spatial patterns in seismic data. They can learn direct mappings from seismic gathers to reservoir properties, bypassing intermediate steps

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and potentially improving accuracy (Downton and Hampson, 2018). However, deep networks typically require large labelled datasets, which are often unavailable in subsurface applications.

Rock Physics Modeling and Synthetic Data

To address data scarcity, synthetic well catalogs can be generated using rock physics models (RPMs). These models simulate elastic responses by varying geological parameters such as porosity, lithology, and saturation, based on real well data (Downton et al., 2022). Seismic responses for these pseudo-wells are modelled using Zoeppritz equations or similar forward models, producing synthetic angle gathers suitable for CNN training.

Transfer Learning

Transfer learning is a technique that allows a model trained on one dataset (synthetic) to be fine-tuned on another (real), thereby combining generalization capability with field-specific adaptation. In this workflow, the CNN is first trained on the synthetic dataset and later updated using a small number of real wells and seismic data to account for field noise and measurement inconsistencies (Downton et al., 2022).

Uncertainty Estimation and Jackknife Validation

Quantifying prediction uncertainty is essential for risk-informed decision-making. The jackknife validation technique addresses this need by systematically leaving one well out of the training dataset in each iteration. This enables the creation of multiple model realizations, from which statistical measures – such as the mean and standard deviation – can be derived, providing spatial confidence maps alongside property predictions (Downton et al., 2022).

Case Study

The study was conducted in the Amberjack Field, located in Mississippi Canyon Block 109, Gulf of Mexico. Geologically, this area represents a complex shelf-edge delta system from the middle Pliocene, consisting of two vertically stacked deltaic sequences separated by a 500-foot (152 m) mud-dominated interval. Both the upper and lower delta sequences are approximately 400 feet (122 m) thick and are composed of progradational clinoforms, delta front slumps, and sandy mouth bars (Mayall et al., 1992).

Hydrocarbons in this field are trapped within five reservoir zones. This study focuses on the Upper Wilcox sand zone, which consists of overlapping sandy clinoforms deposited in a shelf-margin deltaic setting. The Upper Wilcox sand interval presents a mix of stratigraphic and structural complexities and is known to exhibit variable sand distribution, partial fault control, and lateral facies variability. These characteristics require a robust approach to accurately model and predict elastic and petrophysical properties across the field.

Given the depositional environment of the Upper Wilcox sand zone, which includes both unconsolidated and moderately cemented sandstones interbedded with finer-grained sediments, a suitable rock physics model (RPM) was essential. To address this, a sandstone-based RPM incorporating the Matrix Stiffness Index (MSI) was selected, following the methodology of Avseth et al. (2005). This approach accounts for varying degrees of compaction and cementation and allows for effective calibration of elastic responses in transitional lithologies typical for shelf-margin deltaic settings.

Based on the geological setting of the Upper Wilcox sand interval and the calibrated rock physics model, the proposed methodology consists of three main technical components: (1) rock physics modeling, (2) scenario simulation and synthetic data generation, and (3) machine learning model training with hyperparameter tuning and validation.

First, a rock physics model was built and calibrated using four real wells from the Gulf of Mexico study area. The model establishes the relationship between petrophysical properties (such as porosity) and elastic parameters (such as acoustic impedance). Calibration was conducted specifically within the Upper

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Wilcox sand zone, defined between the Top and Base markers, using available well logs and petrophysical interpretation. The calibration result for one of the wells (Well 1) is shown in Figure 1.

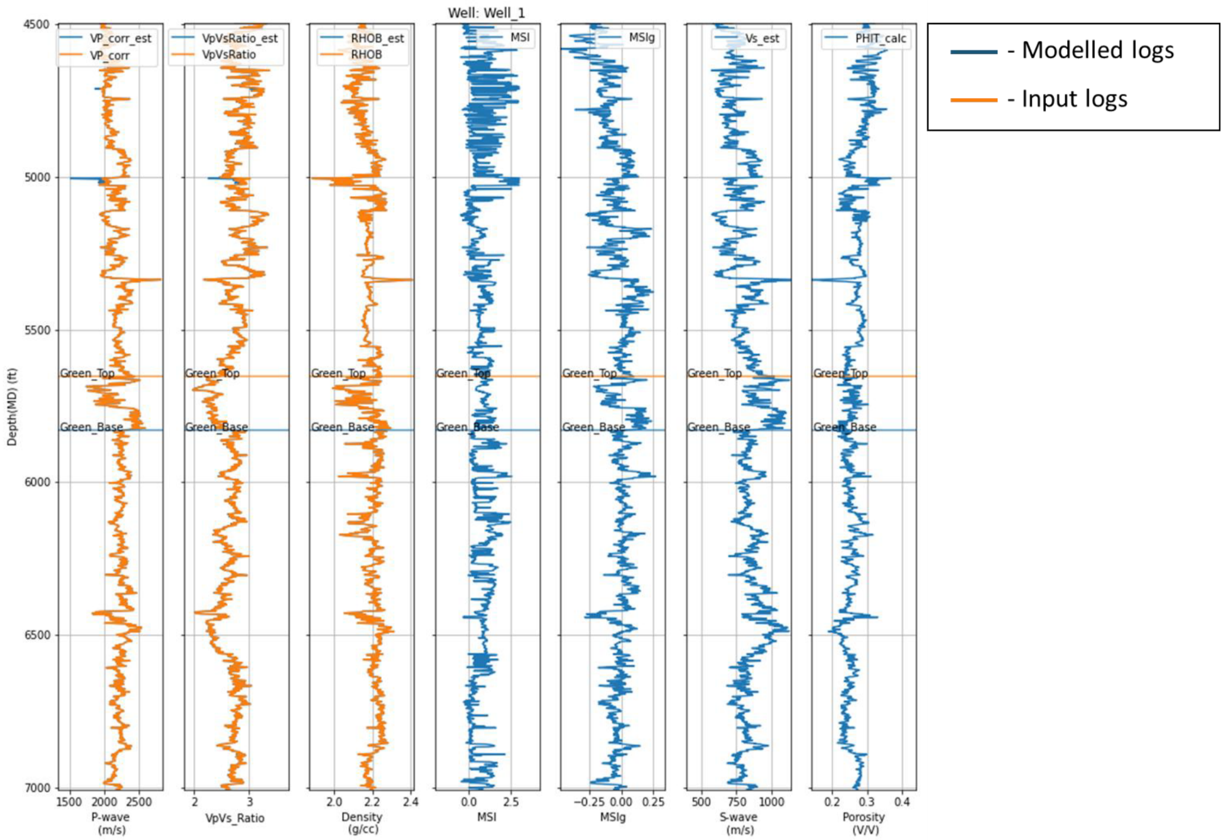

Figure 1—Calibration of the rock physics model (RPM) for Well 1. Input logs (orange) are compared with modelled logs generated using the RPM (blue). The calibration focuses on the Upper Wilcox sand zone interval, bounded by the Top and Base markers.

Input logs (orange) are compared with modelled logs generated using the rock physics model (blue). A good match is observed for the Vp, Vp/Vs ratio, and Density curves within the Upper Wilcox sand interval, indicating that the selected rock physics model accurately captures the elastic behaviour of the reservoir rocks. This agreement confirms the validity of the calibrated model and its suitability for generating realistic synthetic data. A reliable fit at this stage is crucial for ensuring consistency between the simulated pseudo-wells and real data, particularly when the model is later used to create a synthetic catalog.

The model was then perturbed using a set of physically plausible geological scenarios to generate a diverse suite of synthetic pseudo-wells. Scenario design played a critical role in capturing realistic subsurface variability while maintaining consistency with the regional stratigraphy and sedimentological interpretation of the Upper Wilcox sand zone.

To reflect natural heterogeneity, specific reservoir properties were systematically varied.

In the overburden section, only layer thickness was modified using three discrete scaling factors: −10%, 0%, and +10%. These variations allowed the generation of different seismic responses above the target zone, thereby improving the model's ability to generalize to real data affected by overburden interference.

In the target zone, three key parameters were perturbed:

1. Layer thickness, using the same three-point scaling approach as in the overburden, to account for structural and stratigraphic variations in clinoform stacking.

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2. Porosity, varied by ±5% around the base model value, reflecting plausible compaction and depositional variability in deltaic sandstones.
3. Clay volume, discretized into three representative cases: 0 (clean sandstone), 0.5 (moderate shale content), and 1.1 (dominantly shaly). These values were chosen to reflect a full spectrum of lithofacies within the Upper Wilcox sand interval, including both reservoir-quality sands and less favorable interbedded muds.

This scenario matrix, summarized in Table 1, resulted in a robust and well-balanced dataset of 324 synthetic pseudo-wells, encompassing the expected geological variability in the study area. Each pseudo-well was then used to generate synthetic seismic angle gathers. Seismic synthetic modeling was performed using Zoeppritz equations for angle-dependent reflectivity by convolution with a wavelet extracted from real seismic gathers to match the frequency content and phase characteristics of the real data.

Table 1—Scenario matrix used for generating synthetic pseudo-wells. Reservoir and overburden properties were systematically varied to simulate geological uncertainty within the target zone (Upper Wilcox sand interval) and surrounding layers.

Reservoir property		Scenarios
Overburden	Layer thickness	[-10%, 0%, +10%]
Target zone	Layer thickness	[-10%, 0%, +10%]
	Porosity	[-5%, 0%, +5%]
	Clay Volume	[0, 0.5, 1.1]

These synthetic datasets were used to train a convolutional neural network (CNN) to predict three target reservoir properties: P-impedance, porosity, and water saturation. The CNN was designed to capture the nonlinear relationships between seismic input features and these petrophysical outputs.

A variety of CNN hyperparameters were tested during the model development phase, including filter size, number of convolutional layers, dropout rate, number of training epochs, and hidden layer dimensions. The final architecture was selected based on its performance on a held-out validation set, evaluated using standard metrics such as normalized correlation (NCorr), root mean square error (RMSE), and normalized mean square error (NMSE). Training was guided by analyzing learning curves for both training and validation loss, with early stopping used to prevent overfitting.

Figure 2 shows the prediction performance of the trained CNN on one representative synthetic pseudo-well. The comparison includes the input logs, the CNN-predicted logs, and low-frequency model logs.

- For P-impedance, the CNN achieved a normalized correlation of 0.8311, with an RMSE of 158.48 and NMSE of 0.3709, indicating a very strong match across the interval of interest. Although the RMSE value may appear high at first glance, it corresponds to only approximately 3.5% of the average P-impedance value. This relative scale should be kept in mind throughout the document when interpreting non-normalized error values.
- The porosity (PHIT) prediction yielded a normalized correlation of 0.7738, with an RMSE of 0.0101 and NMSE of 0.4851, demonstrating accurate prediction of both trend and magnitude.
- Water saturation (Sw) results were slightly lower in correlation (0.7051), with an RMSE of 0.0702 and NMSE of 0.5945, but still within acceptable limits for synthetic training data, especially considering the sharper transitions in the Sw profile.

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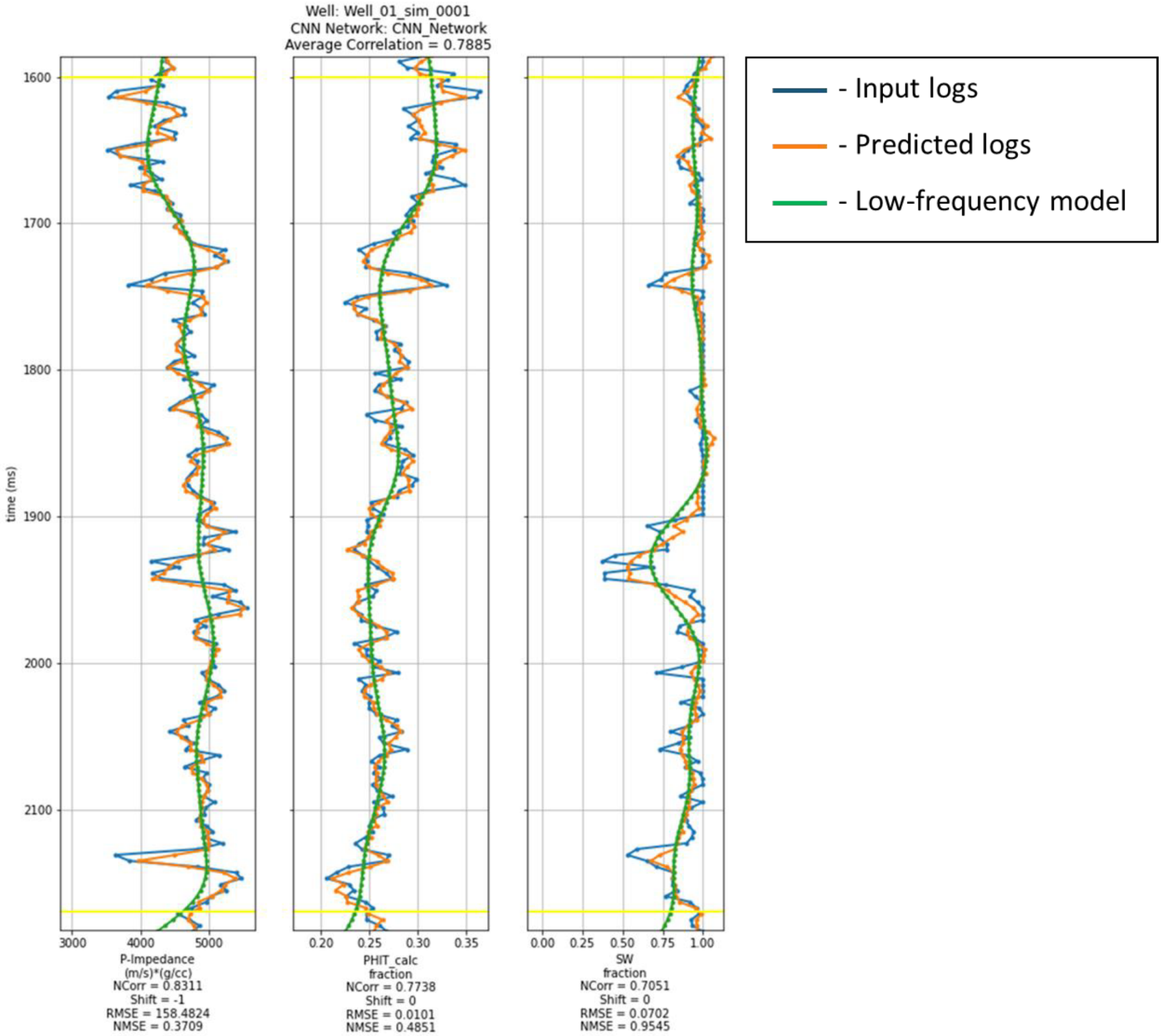

Figure 2—CNN prediction results for synthetic pseudo-well. Comparison of input logs (orange), CNN-predicted logs (blue), and low-frequency model logs (green) for P-impedance, porosity (PHIT), and water saturation (Sw). Performance metrics for each curve are shown below the tracks.

Overall, the average correlation across the three predicted properties was 0.7885, which is considered high for multiparameter prediction tasks. These results confirm that the CNN successfully learned the mapping between synthetic seismic inputs and petrophysical properties, forming a solid foundation for subsequent application to real field data via transfer learning.

After initial training on synthetic data, transfer learning was applied to incorporate real data into the CNN model. In this step, the convolutional layers – responsible for feature extraction – were frozen to preserve their generalized learning from the synthetic dataset. The fully connected layers were then retrained using seismic gathers and logs from four real wells in the study area. This adaptation process allowed the model to fine-tune its predictions based on real seismic characteristics, such as noise, phase variations, and bandwidth differences, which were not present in the synthetic training data.

Figure 3 shows the prediction results of the CNN after transfer learning for one of the real wells (Well 1), focusing on the Upper Wilcox sand interval between Top and Base of the reservoir. The comparison includes input logs (blue), CNN predictions (orange), and low-frequency model logs (green).

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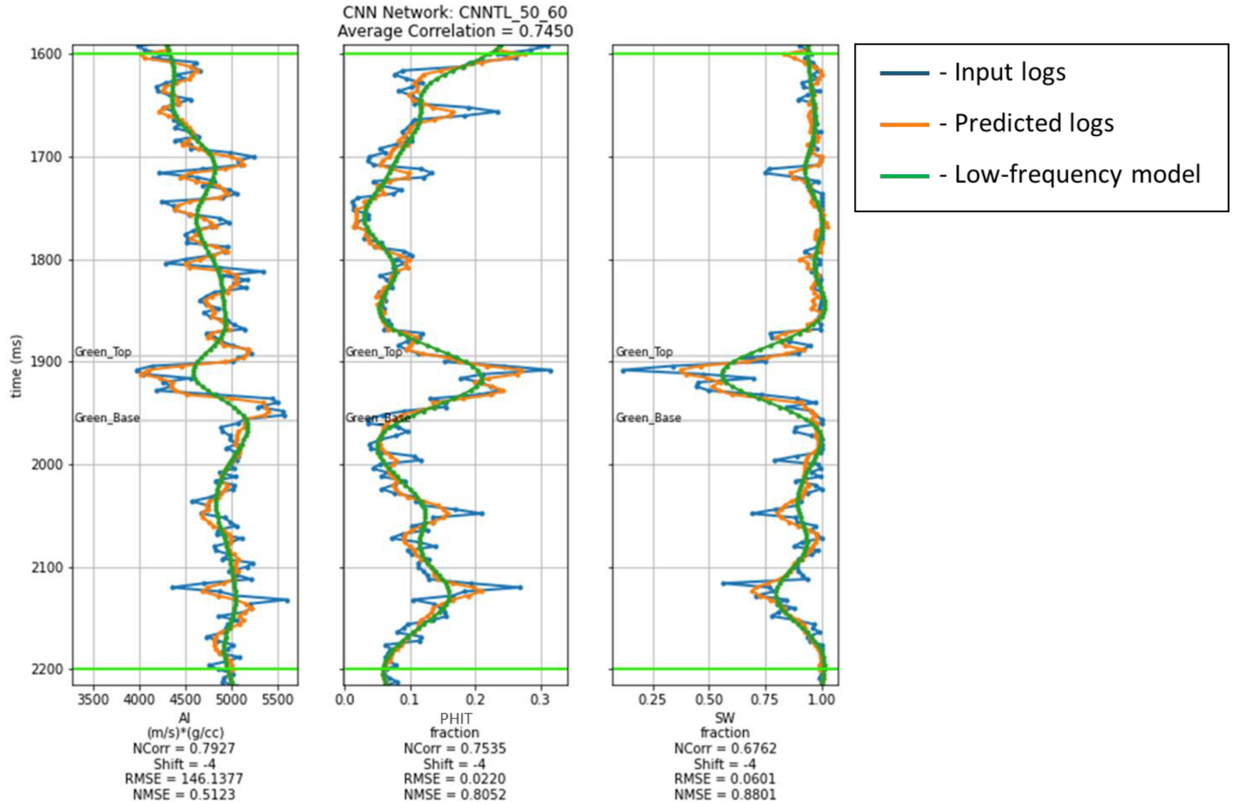

Figure 3—CNN prediction results for real well (Well 1) after transfer learning. Comparison of input logs (blue), CNN predictions (orange), and low-frequency model logs (green) for acoustic impedance (AI), porosity (PHIT), and water saturation (Sw). Performance metrics are shown below each track.

Performance metrics for this well are as follows:

- For Acoustic Impedance (AI): Normalized correlation = 0.7927, RMSE = 146.14, NMSE = 0.5123.
- For Porosity (PHIT): Normalized correlation = 0.7535, RMSE = 0.0220, NMSE = 0.8052.
- For Water Saturation (Sw): Normalized correlation = 0.6762, RMSE = 0.0601, NMSE = 0.8801.

The model performed well in predicting AI and porosity, capturing both the trends and the amplitudes. The Sw prediction, while slightly lower in correlation, still shows consistent alignment with the true log. The lower performance on Sw is expected, given the typically higher variability and nonlinear nature of saturation distribution, as well as its sensitivity to seismic noise.

An average correlation of 0.7450 across all three properties demonstrates that transfer learning effectively improved the model's ability to handle real data, enabling generalization from synthetic simulations to actual field conditions.

The model, after being adjusted to real data using transfer learning, was applied to the entire seismic volume to predict three key reservoir properties across the Upper Wilcox sand zone: P-impedance, porosity, and water saturation. These predictions provided continuous 3D volumes that help better understand the distribution of reservoir qualities in the field.

To evaluate how reliable these predictions are, we used a method called jackknife validation. Since we had four wells available, we ran the workflow multiple times – each time leaving out one well from training and using the remaining three. This way, we created four different prediction models, each trained on slightly different data.

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Each of these models was then applied to the whole study area to produce a full set of property volumes. After that, we compared the results of all four runs. By calculating the average and standard deviation at each location in the 3D volume, we could not only estimate the most likely value of each property, but also how much that value might vary.

The average volumes show the expected property distribution, while the standard deviation volumes show the uncertainty in the predictions. Low uncertainty means the model gave similar results no matter which well was left out – so we can trust those predictions more. High uncertainty may point to areas with noisy seismic data, rapid geological changes, or simply a lack of well control.

This approach gives both the predicted values and a clear picture of how confident we can be in those predictions. That helps make better, more informed decisions when planning wells or evaluating the reservoir.

Presentation of Data and Results

The CNN-generated property volumes closely matched the filtered log curves from real wells, including the blind well (Well 5). This serves as an important validation step, demonstrating the model's ability to generalize beyond the data it was directly trained on.

Figure 4 shows vertical sections through the predicted volumes of P-impedance (top), porosity (middle), and water saturation (bottom) along a line crossing all five wells in the study area. The Green interval is clearly outlined by dashed black horizons, and the well positions are marked for reference.

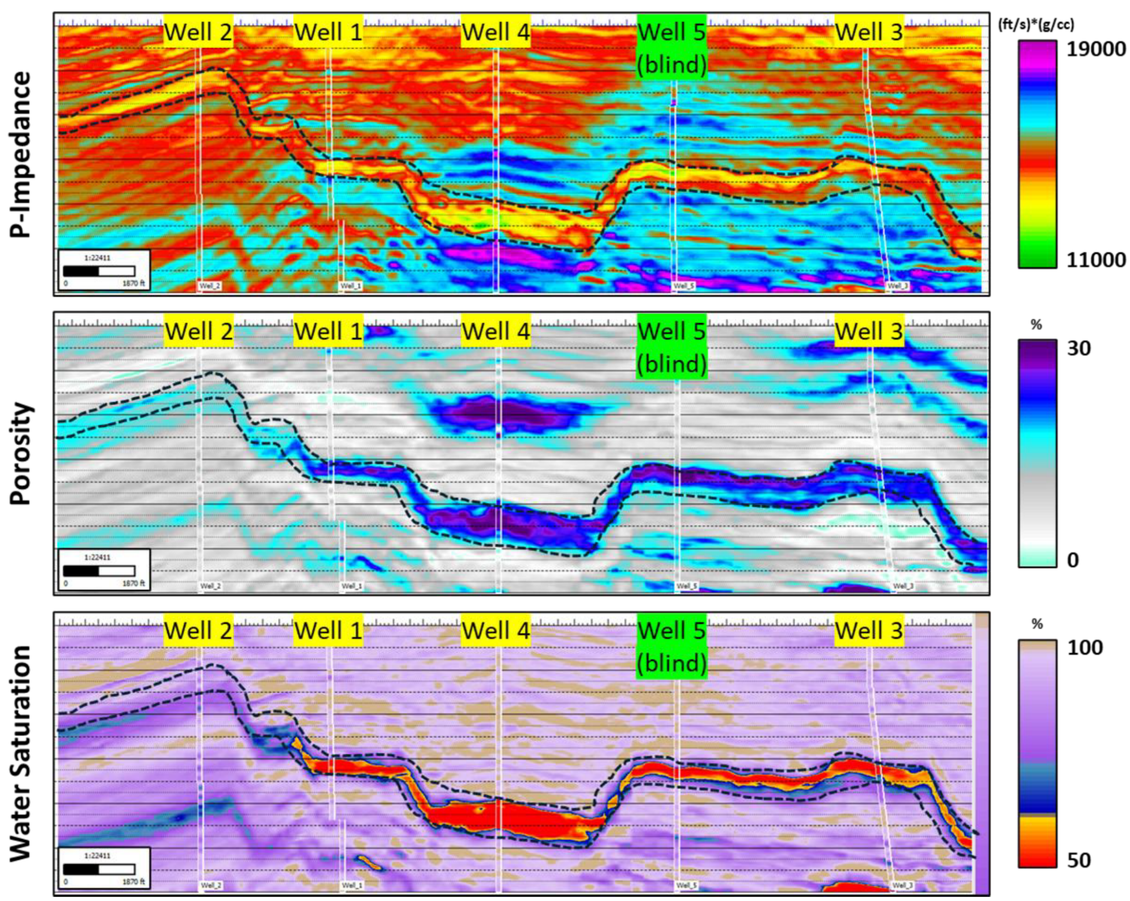

Figure 4—Vertical sections of CNN-predicted volumes along a line crossing all five wells. From top to bottom: P-impedance, porosity, and water saturation. The Upper Wilcox sand interval is marked between Top and Base horizons. Well 5 was used for a blind test, yet the predictions align well with the log data, demonstrating the model's generalization capability.

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The P-impedance section (top) shows strong lateral continuity of impedance contrasts within the Upper Wilcox sand zone, with values that match well control across all wells. Notably, the model was able to correctly capture the structural geometry and impedance trends at Well 5 (blind well).

In the porosity section (middle), zones of higher porosity (blue and cyan colors) are clearly observed within the target interval, particularly between Well 1 and Well 5. These zones correlate with the expected reservoir-quality sandstones seen in well data. The model also reproduces finer porosity variations that align well with changes in seismic character.

The water saturation section (bottom) shows a high degree of differentiation between likely hydrocarbon-bearing zones (lower Sw, shown in warm colors) and water-saturated regions (higher Sw, in blue). Areas of low water saturation are consistent with those of higher porosity, supporting the physical realism of the prediction.

Overall, the figure confirms that the CNN not only preserves the main trends observed in the well data but also captures lateral variations across the field. The strong match at the blind well provides additional confidence that the model is not overfitted and can be used reliably in nearby areas with limited well control.

To estimate prediction uncertainty, the jackknife technique was used to generate multiple realizations of the CNN model – each trained with a different combination of three out of the four available wells. For each reservoir property (P-impedance, porosity, and water saturation), four property volumes were created. These were then used to calculate the mean and standard deviation volumes, which reflect the average predicted value and the variation across models, respectively.

Figure 5 presents horizontal maps through the Upper Wilcox sand zone, showing both the mean predictions (top row) and the corresponding standard deviation maps (bottom row) for the three predicted properties: P-impedance (left), porosity (center), and water saturation (right). These maps help evaluate not just where good reservoir properties occur, but also how confident the model is in those predictions.

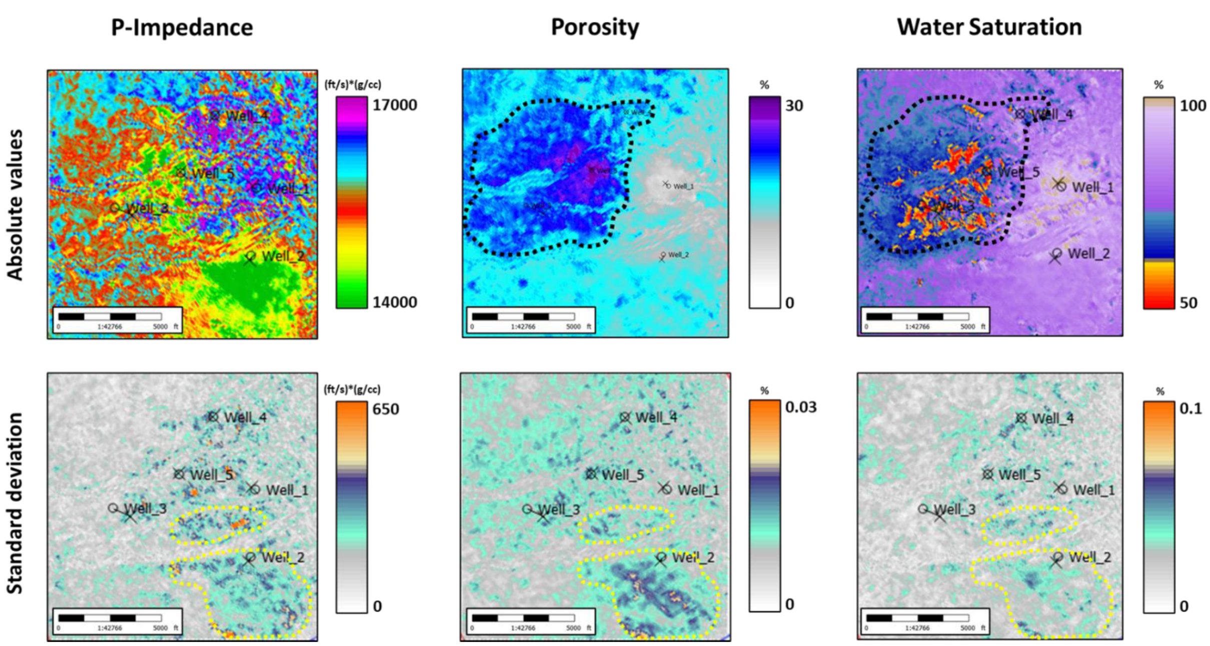

Figure 5—Horizontal property and uncertainty maps for the Upper Wilcox sand zone. Top row: predicted mean values for P-impedance, porosity, and water saturation. Bottom row: standard deviation maps showing prediction uncertainty. Dotted contours highlight key zones with high property quality and low uncertainty, indicating reliable and geologically meaningful predictions.

In the porosity map, a clearly defined zone in the central to northwestern area shows high porosity values (blue colours) with low associated uncertainty in the standard deviation map below. This combination indicates a region where the model consistently predicts favourable reservoir characteristics regardless of

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which well is excluded, suggesting high confidence and high quality of prediction. This area also aligns well with Well 5 (blind well), reinforcing the validity of the result.

In contrast, the southeastern part of the volume displays higher standard deviation, particularly in the water saturation map. This suggests increased prediction uncertainty, likely caused by factors such as seismic noise, complex stratigraphy, or lack of nearby well control. Such zones should be interpreted with caution, as the variability in model outputs points to less stable predictions.

The grey areas in the standard deviation maps mark regions with very low uncertainty, implying stable model behaviour across all jackknife realizations. These areas support the reliability of the CNN predictions and can be considered as more robust targets for further reservoir evaluation or drilling decisions.

Conclusions

This study demonstrates a robust methodology for predicting reservoir properties and quantifying associated uncertainties using a machine learning framework guided by rock physics. By integrating synthetic data generation, transfer learning, and jackknife-based uncertainty estimation, the approach addresses two of the most common limitations in subsurface prediction: limited well control and the lack of confidence measures in ML outputs.

The use of a rock physics-calibrated synthetic well catalog proved particularly effective for training deep learning models in a data-scarce environment. This enabled the CNN to learn from a broader spectrum of geological scenarios than would be possible with real wells alone. Transfer learning further allowed the model to adapt to the specific noise characteristics and resolution of the real seismic dataset, improving predictive accuracy and well ties.

The application of the jackknife technique added an essential layer of statistical reliability to the predictions. Instead of providing a single deterministic outcome, the method delivered multiple realizations that allowed the generation of confidence maps. These maps offer valuable insights into spatial variability and prediction stability which is an important asset for reservoir risk assessment and development planning.

From an operational perspective, the approach is scalable and adaptable to other fields with limited well coverage or complex geology. It reduces reliance on conventional inversion workflows and offers a more direct, data-driven alternative with built-in uncertainty quantification. The identification of areas with both favorable property values and low uncertainty is particularly useful for targeting high-confidence zones for drilling or further evaluation.

Overall, this study highlights the potential of combining domain knowledge with machine learning to enhance reservoir characterization in challenging exploration settings. Future work may expand this framework to include additional target properties, more advanced uncertainty quantification techniques, and real-time decision support for reservoir development.

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